REG.NO.: 26MA10076

SLOT: C2 + TC2

VIT
Vellore Institute of Technology

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
FALL SEMESTER 2026-2027

Programme Name & Branch	: M.Tech CSE & M.Tech CSE (AI and ML)
Course Code and Course Name	: MACSE512 & Operating Systems
Faculty Name(s)	: Prof. Ananda Kumar S & Prof. Vijayasherly V
Class Number(s)	: VL2026270106059, VL2026270106049
Date of Examination	: 11/08/2026
Exam Duration	: 90 minutes

Maximum Marks: 50

General instruction(s):

- Answer All Questions
- Course Outcomes:
  1. Explain the structure and key components of an operating system including system calls, interfaces, and virtualization techniques.
  2. Analyze process lifecycle, scheduling techniques, and implement synchronization mechanisms in multi-threaded environments.

Q. No	Question	M	CO	BL
1.	a) Suppose that you are asked to design a mini operating system for a hand-held device, illustrate the major modules using a suitable architectural diagram, and justify your design choices. [5 marks] b) Explain the key activities of an operating system with respect to Memory Management and File management. [5 marks]	10	1	2
2.	What is a system call? Why are system calls necessary? How would you differentiate between a system call and a system program? Describe them in detail with appropriate examples.	10	1	2
3	a) How do we represent the context of a process? Describe the data structures associated with it. [5 Marks] b) Assume that you are asked to develop a web-based application that must be deployed on three different operating systems. The development team wants to avoid maintaining separate versions of the application for each operating system while ensuring consistent execution across all platforms. Design a suitable system architecture that enables the same application to run on all three operating systems. Draw the architecture diagram and explain the role of each component in your design. [5 Marks]	10	1	3
4.	a) Calculate average turnaround time and average waiting time for Round Robin (RR, time quantum=2 ms) and Shortest-Remaining Time First (SRTF) CPU scheduling algorithms with Gantt charts for the data given below. [6 Marks]	10	2	3

Process ID	Arrival Time (ms)	Burst Time (ms)
A	1	9
B	3	4
C	2	6
D	6	8
E	4	5

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b) A computer system uses Multilevel Queue CPU Scheduling with three queues arranged in descending order of priority.

- Q1 (Highest Priority): Preemptive Priority Scheduling
- Q2: First Come First Served (FCFS)
- Q3 (Lowest Priority): Round Robin (Time Quantum = 2 ms)

Processes are permanently assigned to a queue and cannot move between queues. The scheduler always executes the highest-priority non-empty queue. Scheduling is preemptive between queues. The process details are given below:

Process	Arrival Time (ms)	Burst Time (ms)	Priority	Queue
P1	0	6	2	Q2
P2	1	4	1	Q1
P3	2	5	3	Q3
P4	3	3	2	Q1
P5	4	2	1	Q2

Draw the Gantt Chart for the given processes using Multilevel Queue Scheduling. Calculate the Waiting Time (WT) and Turnaround Time (TAT) for each process.
[4 Marks]

5. A cloud service provider is running six Virtual Machines (VMs) that share three categories of resources. The resource allocation snapshot is shown below.

VM	Allocation	Outstanding Request
P_0	1 0 1	0 1 0
P_1	2 1 0	2 0 1
P_2	3 1 2	0 0 0
P_3	2 1 1	1 2 0
P_4	1 0 2	1 1 1
P_5	0 1 0	1 1 0

Available Resources = (1,0,0)

The cloud administrator suspects that some virtual machines may be waiting indefinitely for resources.

Analyze the current system state and determine whether the administrator's concern is valid. Support your answer by tracing the execution of the appropriate operating system algorithm with step-wise results.

10 | 2 | 3

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